gpeyt1-36 · Rankine thrust with water table

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 22 ft high vertical-backed wall retains level granular backfill (c = 0, φ = 32°). Because the weep holes are clogged, a water table has built up 8 ft below the top of the backfill. γ = 118 pcf above the water table, γsat = 126 pcf below. Ignore wall friction; the wall moves enough to mobilize active pressure. The concrete stem is 18 in. thick.

Most nearly, the total horizontal force (soil + water) per foot of wall is:

A. 7,136 lb/ft
B. 13,251 lb/ft
C. 9,015 lb/ft
D. 8,774 lb/ft

Given equations / data

> $K_a = \tan^2(45 - \varphi/2)$; $\sigma'_h = K_a \sigma'_v$; $u = \gamma_w h_w$; P = area of pressure diagram

Worked solution

1. $K_a = 0.3073$
2. σ'_h at 8 ft = $0.3073 \times 118 \times 8 = 290$ psf
3. σ'_h at 22 ft = $290 + 0.3073 \times 63.6 \times 14 = 564$ psf
4. Soil: $\frac{1}{2}(290)(8) + 290(14) + \frac{1}{2}(274)(14) = 7{,}136$ lb/ft
5. Water: $\frac{1}{2}(62.4)(14)^2 = 6{,}115$ lb/ft
6. Total = 13,251 lb/ft

Final answer

13,251 lb/ft

Why the other values are wrong

- **7,136 lb/ft** — forgot the water pressure
- **9,015 lb/ft** — used γsat (total) below the WT and no separate water
- **8,774 lb/ft** — treated the backfill as dry

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine Earth Coefficients, p. 72

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=w48hBSXrUhY (no creator values copied).

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